

```
lahari@Lahari:~$ pwd
/home/lahari
lahari@Lahari:~$ gcc --version
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0
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warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

lahari@Lahari:~$ mkdir OS_Skill1
lahari@Lahari:~$ cd OS_Skill1
lahari@Lahari:~/OS_Skill1$ pwd
/home/lahari/OS_Skill1
lahari@Lahari:~/OS_Skill1$ ls
lahari@Lahari:~/OS_Skill1$ nano hello.c
lahari@Lahari:~/OS_Skill1$ cat hello.c
#include <stdio.h>

int main()
{
    printf("Hello Linux!\n");
    return 0;
}

lahari@Lahari:~/OS_Skill1$ gcc hello.c -o hello
lahari@Lahari:~/OS_Skill1$ ./hello
Hello Linux!
```

```
lahari@Lahari:~/OS_Skill1$ ls
hello hello.c
lahari@Lahari:~/OS_Skill1$ pwd
/home/lahari/OS_Skill1
lahari@Lahari:~/OS_Skill1$ ls
hello hello.c
lahari@Lahari:~/OS_Skill1$ cd ..
lahari@Lahari:~$ pwd
/home/lahari
lahari@Lahari:~$ cd OS_Skill1
lahari@Lahari:~/OS_Skill1$ pwd
/home/lahari/OS_Skill1
lahari@Lahari:~/OS_Skill1$ mkdir testfolder
lahari@Lahari:~/OS_Skill1$ ls
hello hello.c testfolder
lahari@Lahari:~/OS_Skill1$ rmdir testfolder
lahari@Lahari:~/OS_Skill1$ ls
hello hello.c
lahari@Lahari:~/OS_Skill1$ echo "Operating Systems Skill Lab" > sample.txt
lahari@Lahari:~/OS_Skill1$ ls
hello hello.c sample.txt
lahari@Lahari:~/OS_Skill1$ cat sample.txt
Operating Systems Skill Lab
lahari@Lahari:~/OS_Skill1$ cp sample.txt sample_copy.txt
lahari@Lahari:~/OS_Skill1$ ls
hello hello.c sample.txt sample_copy.txt
lahari@Lahari:~/OS_Skill1$ mv sample_copy.txt renamed.txt
lahari@Lahari:~/OS_Skill1$ ls
hello hello.c renamed.txt sample.txt
lahari@Lahari:~/OS_Skill1$ cat renamed.txt
Operating Systems Skill Lab
lahari@Lahari:~/OS_Skill1$ find . -name "renamed.txt"
./renamed.txt
lahari@Lahari:~/OS_Skill1$ rm renamed.txt
lahari@Lahari:~/OS_Skill1$ ls
hello hello.c sample.txt
lahari@Lahari:~/OS_Skill1$ find . -name "renamed.txt"
lahari@Lahari:~/OS_Skill1$ history
 1 sudo apt update
 2 sudo apt install build-essential
 3 gcc --version
 4 mkdir OSLab
 5 ~/OSLab$
 6 uname -a
 7 sudo apt install <deb name>
 8 uname -a
```

```
7 sudo apt install <deb name>
8 uname -a
9 lscpu
10 lsblk
11 lsblk
12 ps -ef
13 top observation
14 top
15 q
16 lscpu
17 free -h
18 lsblk
19 #include <stdio.h>
20 gcc --version
21 nano program1.c
22 username -a
23 nano program1.c
24 cd ~/Downloads
25 nano fork1.c
26 ls
27 gcc fork1.c -o fork1
28 ./fork1
29 cd ~/Downloads
30 ls
31 gcc fork1.c -o fork1
32 gcc --version
33 ./fork1
34 nano program1.c
35 gcc program1.c -o program1
36 ./program1
37 nano program2.c
38 cd ~/Downloads
39 ls
40 cd OSLab1
41 pwd
42 echo $SHELL
43 cd /mnt/c/Users/surya/Desktop/OSLab1
44 pwd
45 nano program1.c
46 gcc program1.c -o program1
47 ./program1
48 cd /mnt/c/Users/surya/Desktop
49 ls
50 cd /mnt/c/Users
51 ls
52 cd /mnt/c/Users/surya
```

```
272 history
lahari@Lahari:~/OS_Skill1$ echo $SHELL
/bin/bash
lahari@Lahari:~/OS_Skill1$ bash --version
GNU bash, version 5.3.9(1)-release (x86_64-pc-linux-gnu)
Copyright (C) 2025 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html
>

This is free software; you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
lahari@Lahari:~/OS_Skill1$ pwd
/home/lahari/OS_Skill1
lahari@Lahari:~/OS_Skill1$ git --version
git version 2.53.0
lahari@Lahari:~/OS_Skill1$ sudo apt update
[sudo: authenticate] Password:
sudo: Authentication failed - try again
```

```
lahari@Lahari:~/OS_Skill1$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    hello
    hello.c
    sample.txt

nothing added to commit but untracked files present (use "git add" to track)
lahari@Lahari:~/OS_Skill1$ mkdir src include bin docs
lahari@Lahari:~/OS_Skill1$ ls
bin docs hello hello.c include sample.txt src
lahari@Lahari:~/OS_Skill1$ mv hello.c src/
lahari@Lahari:~/OS_Skill1$ ls
bin docs hello include sample.txt src
lahari@Lahari:~/OS_Skill1$ ls src
hello.c
lahari@Lahari:~/OS_Skill1$ cat src/hello.c
#include <stdio.h>

int main()
{
    printf("Hello Linux!\n");
    return 0;
}

lahari@Lahari:~/OS_Skill1$ mv hello bin/
lahari@Lahari:~/OS_Skill1$ ls
bin docs include sample.txt src
lahari@Lahari:~/OS_Skill1$ ls bin
hello
lahari@Lahari:~/OS_Skill1$ ./bin/hello
Hello Linux!
lahari@Lahari:~/OS_Skill1$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    bin/
```

```

lahari@Lahari:~/OS_Skill1$ git add .
lahari@Lahari:~/OS_Skill1$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   bin/hello
        new file:   sample.txt
        new file:   src/hello.c

lahari@Lahari:~/OS_Skill1$ git config --global user.name "Lahari"
lahari@Lahari:~/OS_Skill1$ git config --global user.email "lahari2007.namala@gmail.com"
lahari@Lahari:~/OS_Skill1$ git config --global user.name Lahari
lahari@Lahari:~/OS_Skill1$ git config --global user.email lahari2007.namala@gmail.com
lahari@Lahari:~/OS_Skill1$ git config --global user.email lahari2007.namala@gmail.com
lahari@Lahari:~/OS_Skill1$ git commit -m "Initial OS Skill 1 project"
[master (root-commit) 0791805] Initial OS Skill 1 project
 3 files changed, 9 insertions(+)
 create mode 100755 bin/hello
 create mode 100644 sample.txt
 create mode 100644 src/hello.c
lahari@Lahari:~/OS_Skill1$ echo $SHELL
/bin/bash
lahari@Lahari:~/OS_Skill1$ uname -r
6.18.33.2-microsoft-standard-WSL2
lahari@Lahari:~/OS_Skill1$ uname -a
Linux Lahari 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 GNU/Linux
lahari@Lahari:~/OS_Skill1$ nano Makefile
lahari@Lahari:~/OS_Skill1$ cat Makefile
CC=gcc
CFLAGS=-Wall

all:
    $(CC) $(CFLAGS) src/hello.c -o bin/hello

clean:
    rm -f bin/hello
lahari@Lahari:~/OS_Skill1$ make

```



```
lahari@Lahari:~/OS_Skill1$ make
gcc -Wall src/hello.c -o bin/hello
lahari@Lahari:~/OS_Skill1$ ./bin/hello
Hello Linux!
lahari@Lahari:~/OS_Skill1$ make clean
rm -f bin/hello
lahari@Lahari:~/OS_Skill1$ ls bin
lahari@Lahari:~/OS_Skill1$ make
gcc -Wall src/hello.c -o bin/hello
lahari@Lahari:~/OS_Skill1$ ls bin
hello
lahari@Lahari:~/OS_Skill1$ nano src/fork_demo.c
lahari@Lahari:~/OS_Skill1$ cat src/fork_demo.c
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>

int main()
{
    pid_t pid;

    printf("Before fork: PID = %d\n", getpid());

    pid = fork();

    if (pid < 0)
    {
        printf("Fork failed\n");
    }
    else if (pid == 0)
    {
        printf("Child Process\n");
        printf("Child PID = %d\n", getpid());
        printf("Parent PID = %d\n", getppid());
    }
    else
    {
        printf("Parent Process\n");
        printf("Parent PID = %d\n", getpid());
        printf("Child PID = %d\n", pid);
    }

    return 0;
}
lahari@Lahari:~/OS_Skill1$ gcc src/fork_demo.c -o bin/fork_demo
lahari@Lahari:~/OS_Skill1$ ./bin/fork_demo
```

```

lahari@Lahari:~/OS_Skill1$ gcc src/fork_demo.c -o bin/fork_demo
lahari@Lahari:~/OS_Skill1$ ./bin/fork_demo
Before fork: PID = 1496
Parent Process
Parent PID = 1496
Child PID = 1497
Child Process
Child PID = 1497
Parent PID = 330
lahari@Lahari:~/OS_Skill1$ ./bin/fork_demo
Before fork: PID = 1507
Parent Process
Parent PID = 1507
Child PID = 1508
Child Process
Child PID = 1508
Parent PID = 1507
lahari@Lahari:~/OS_Skill1$ nano src/multiple_fork.c
lahari@Lahari:~/OS_Skill1$ cat src/multiple_fork.c
#include <stdio.h>
#include <unistd.h>

int main()
{
    printf("Original Process PID = %d\n", getpid());

    fork();
    fork();

    printf("PID = %d, PPID = %d\n", getpid(), getppid());

    return 0;
}
lahari@Lahari:~/OS_Skill1$ gcc src/multiple_fork.c -o bin/multiple_fork
lahari@Lahari:~/OS_Skill1$ ./bin/multiple_fork
Original Process PID = 1554
PID = 1554, PPID = 334
PID = 1556, PPID = 330
PID = 1555, PPID = 330
PID = 1557, PPID = 1555
lahari@Lahari:~/OS_Skill1$ nano src/execl_demo.c
lahari@Lahari:~/OS_Skill1$ cat src/execl_demo.c
#include <stdio.h>
#include <unistd.h>

```



```

printf("Before execl\n");

execl("/bin/ls", "ls", "-l", NULL);

printf("This line executes only if execl fails\n");

return 0;
}
lahari@Lahari:~/OS_Skill1$ gcc src/execl_demo.c -o bin/execl_demo
lahari@Lahari:~/OS_Skill1$ ./bin/execl_demo
Before execl
total 24
-rw-r--r-- 1 lahari lahari 93 Aug 12 06:51 Makefile
drwxr-xr-x 2 lahari lahari 4096 Aug 12 06:56 bin
drwxr-xr-x 2 lahari lahari 4096 Aug 12 06:46 docs
drwxr-xr-x 2 lahari lahari 4096 Aug 12 06:46 include
-rw-r--r-- 1 lahari lahari 28 Aug 12 06:41 sample.txt
drwxr-xr-x 2 lahari lahari 4096 Aug 12 06:56 src
lahari@Lahari:~/OS_Skill1$ ls -l
total 24
-rw-r--r-- 1 lahari lahari 93 Aug 12 06:51 Makefile
drwxr-xr-x 2 lahari lahari 4096 Aug 12 06:56 bin
drwxr-xr-x 2 lahari lahari 4096 Aug 12 06:46 docs
drwxr-xr-x 2 lahari lahari 4096 Aug 12 06:46 include
-rw-r--r-- 1 lahari lahari 28 Aug 12 06:41 sample.txt
drwxr-xr-x 2 lahari lahari 4096 Aug 12 06:56 src
lahari@Lahari:~/OS_Skill1$ nano src/execvp_demo.c
lahari@Lahari:~/OS_Skill1$ cat src/execvp_demo.c
#include <stdio.h>
#include <unistd.h>

int main()
{
    char *args[] = {"ls", "-l", NULL};

    printf("Before execvp\n");

    execvp("ls", args);

    printf("execvp failed\n");

    return 0;
}
lahari@Lahari:~/OS_Skill1$ gcc src/execvp_demo.c -o bin/execvp_demo
lahari@Lahari:~/OS_Skill1$ ./bin/execvp_demo
Before execvp

```

```

lahari@Lahari:~/OS_Skill1$ cat src/wait_demo.c
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    pid_t pid = fork();

    if (pid == 0)
    {
        printf("Child started. PID = %d\n", getpid());
        sleep(2);
        printf("Child completed.\n");
    }
    else
    {
        printf("Parent waiting for child...\n");

        wait(NULL);

        printf("Child finished. Parent continuing.\n");
    }

    return 0;
}

lahari@Lahari:~/OS_Skill1$ gcc src/wait_demo.c -o bin/wait_demo
lahari@Lahari:~/OS_Skill1$ ./bin/wait_demo
Parent waiting for child...
Child started. PID = 1770
Child completed.
Child finished. Parent continuing.
lahari@Lahari:~/OS_Skill1$ nano src/waitpid_demo.c
lahari@Lahari:~/OS_Skill1$ cat src/waitpid_demo.c
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    pid_t pid = fork();

    if (pid == 0)
    {
        printf("Child PID = %d\n", getpid());
        sleep(2);
    }
}

```

```

Child completed.
Child finished. Parent continuing.
lahari@Lahari:~/OS_Skill1$ nano src/waitpid_demo.c
lahari@Lahari:~/OS_Skill1$ cat src/waitpid_demo.c
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    pid_t pid = fork();

    if (pid == 0)
    {
        printf("Child PID = %d\n", getpid());
        sleep(2);
        printf("Child completed\n");
    }
    else
    {
        printf("Parent PID = %d\n", getpid());
        printf("Parent waiting for child %d...\n", pid);

        waitpid(pid, NULL, 0);

        printf("Child %d completed. Parent continuing.\n", pid);
    }

    return 0;
}
lahari@Lahari:~/OS_Skill1$ ^C
lahari@Lahari:~/OS_Skill1$ gcc src/waitpid_demo.c -o bin/waitpid_demo
lahari@Lahari:~/OS_Skill1$ ./bin/waitpid_demo
Parent PID = 1819
Parent waiting for child 1820...
Child PID = 1820
Child completed
Child 1820 completed. Parent continuing.
lahari@Lahari:~/OS_Skill1$ nano src/process_tree.c

```

```

lahari@Lahari:~/OS_Skill1$ gcc src/waitpid_demo.c -o bin/waitpid_demo
lahari@Lahari:~/OS_Skill1$ ./bin/waitpid_demo
Parent PID = 1819
Parent waiting for child 1820...
Child PID = 1820
Child completed
Child 1820 completed. Parent continuing.
lahari@Lahari:~/OS_Skill1$ nano src/process_tree.c
lahari@Lahari:~/OS_Skill1$ cat src/process_tree.c
#include <stdio.h>
#include <unistd.h>

int main()
{
    fork();
    fork();

    printf("PID = %d, PPID = %d\n", getpid(), getppid());

    sleep(30);

    return 0;
}
lahari@Lahari:~/OS_Skill1$ gcc src/process_tree.c -o bin/process_tree
lahari@Lahari:~/OS_Skill1$ ./bin/process_tree &
[1] 1870
PID = 1870, PPID = 334
PID = 1874, PPID = 1870
PID = 1872, PPID = 1870
PID = 1875, PPID = 1872
lahari@Lahari:~/OS_Skill1$ ps -o pid,ppid,state,cmd -C process_tree
  PID   PPID  S  CMD
  1870    334  S  ./bin/process_tree
  1872   1870  S  ./bin/process_tree
  1874   1870  S  ./bin/process_tree
  1875   1872  S  ./bin/process_tree
lahari@Lahari:~/OS_Skill1$ pstree -p | grep process_tree
      |
      |--process_tree(1870)--+-process_tree(1872)---proc+
      |
      |                        \--process_tree(1874)
lahari@Lahari:~/OS_Skill1$ ./bin/process_tree &
[2] 1904
PID = 1904, PPID = 334
PID = 1907, PPID = 1904
PID = 1908, PPID = 1906

```

```

lahari@Lahari:~/OS_Skill1$ ./bin/execvp_demo
Before execvp
total 24
-rw-r--r-- 1 lahari lahari 93 Aug 12 06:51 Makefile
drwxr-xr-x 2 lahari lahari 4096 Aug 12 06:58 bin
drwxr-xr-x 2 lahari lahari 4096 Aug 12 06:46 docs
drwxr-xr-x 2 lahari lahari 4096 Aug 12 06:46 include
-rw-r--r-- 1 lahari lahari 28 Aug 12 06:41 sample.txt
drwxr-xr-x 2 lahari lahari 4096 Aug 12 06:57 src
lahari@Lahari:~/OS_Skill1$ nano src/execv_demo.c
lahari@Lahari:~/OS_Skill1$ cat src/execv_demo.c
#include <stdio.h>
#include <unistd.h>

int main()
{
    char *args[] = {"ls", "-l", NULL};

    printf("Before execv\n");

    execv("/bin/ls", args);

    printf("execv failed\n");

    return 0;
}

lahari@Lahari:~/OS_Skill1$ gcc src/execv_demo.c -o bin/execv_demo
lahari@Lahari:~/OS_Skill1$ ./bin/execv_demo
Before execv
total 24
-rw-r--r-- 1 lahari lahari 93 Aug 12 06:51 Makefile
drwxr-xr-x 2 lahari lahari 4096 Aug 12 06:59 bin
drwxr-xr-x 2 lahari lahari 4096 Aug 12 06:46 docs
drwxr-xr-x 2 lahari lahari 4096 Aug 12 06:46 include
-rw-r--r-- 1 lahari lahari 28 Aug 12 06:41 sample.txt
drwxr-xr-x 2 lahari lahari 4096 Aug 12 06:58 src
lahari@Lahari:~/OS_Skill1$ nano src/wait_demo.c
lahari@Lahari:~/OS_Skill1$ cat src/wait_demo.c
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{

```



```

[2] 1904
PID = 1904, PPID = 334
PID = 1907, PPID = 1904
PID = 1908, PPID = 1906
PID = 1906, PPID = 1904
lahari@Lahari:~/OS_Skill1$ strace --version
strace -- version 6.19
Copyright (c) 1991-2026 The strace developers <https://strace.io>.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

Optional features enabled: stack-trace=libunwind stack-demangle m32-mpers mx
32-mpers
lahari@Lahari:~/OS_Skill1$ strace ls
execve("/usr/bin/ls", ["ls"], 0x7ffdd32d9d10 /* 26 vars */) = 0
brk(NULL) = 0x5e1da6dcf000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0
x718aa7213000
access("/etc/ld.so.preload", R_OK) = -1 ENOENT (No such file or directo
ry)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=20087, ...}) = 0
mmap(NULL, 20087, PROT_READ, MAP_PRIVATE, 3, 0) = 0x718aa720e000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libselinux.so.1", O_RDONLY|O_CLO
EXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0"...
, 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=199120, ...}) = 0
mmap(NULL, 206288, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x718aa71db
000
mmap(0x718aa71e2000, 135168, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_
DENYWRITE, 3, 0x7000) = 0x718aa71e2000
mmap(0x718aa7203000, 28672, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE,
3, 0x28000) = 0x718aa7203000
mmap(0x718aa720a000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_D
ENYWRITE, 3, 0x2f000) = 0x718aa720a000
mmap(0x718aa720c000, 5584, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_A
NONYMOUS, -1, 0) = 0x718aa720c000
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libgcc_s.so.1", O_RDONLY|O_CLOEX
EC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0"...
, 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=187120, ...}) = 0
mmap(NULL, 185256, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x718aa71ad

```



```

x82\x88\x3c\x91\x24\x15\x12\xa7\x5b\x3d\x61\x79\x48\xcb\x39\xe1", 32, 0) = 3
2
munmap(0x7826cae38000, 323584) = 0
write(1, "Wed Aug 12 07:03:32 UTC 2026\n", 29Wed Aug 12 07:03:32 UTC 2026
) = 29
sigaltstack({ss_sp=NULL, ss_flags=SS_DISABLE, ss_size=8192}, NULL) = 0
munmap(0x7826cb0c8000, 12288) = 0
exit_group(0) = ?
+++ exited with 0 +++
lahari@Lahari:~/OS_Skill1$ strace -o trace.txt ls
Makefile bin docs include sample.txt src trace.txt
lahari@Lahari:~/OS_Skill1$ ls
Makefile bin docs include sample.txt src trace.txt
lahari@Lahari:~/OS_Skill1$ grep "execve(" trace.txt
execve("/usr/bin/ls", ["ls"], 0x7ffdb9389c10 /* 26 vars */) = 0
lahari@Lahari:~/OS_Skill1$ grep "openat(" trace.txt | head
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libselinux.so.1", O_RDONLY|O_CLO
EXEC) = 3
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libgcc_s.so.1", O_RDONLY|O_CLOEX
EC) = 3
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libm.so.6", O_RDONLY|O_CLOEXEC)
= 3
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC)
= 3
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libpcre2-8.so.0", O_RDONLY|O_CLO
EXEC) = 3
openat(AT_FDCWD, "/proc/filesystems", O_RDONLY|O_CLOEXEC) = 3
openat(AT_FDCWD, "/proc/mounts", O_RDONLY|O_CLOEXEC) = 3
openat(AT_FDCWD, "/proc/self/maps", O_RDONLY|O_CLOEXEC) = 3
openat(AT_FDCWD, "/usr/share/coreutils/locales/uucore/en-US.ftl", O_RDONLY|O
_CLOEXEC) = 3
lahari@Lahari:~/OS_Skill1$ grep "read(" trace.txt | head
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0"...
, 832) = 832
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0"...
, 832) = 832
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0"...
, 832) = 832
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0 \250\2\0\0\0\0"...
, 832) = 832
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0"...
, 832) = 832
read(3, "nodev\tsysfs\nnodev\ttmpfs\nnodev\tbd"... , 1024) = 442
read(3, "none /usr/lib/modules/6.18.33.2-"... , 1024) = 1024
read(3, "ime 0 0\ndepts /dev/pts devpts r"... , 1024) = 1024

```

[illegible]

```
lahari@Lahari:~/OS_Skill1$ nano src/minishell.c
lahari@Lahari:~/OS_Skill1$ cat src/minishell.c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <string.h>
#include <sys/wait.h>

int main()
{
    char command[100];

    while (1)
    {
        printf("MiniShell> ");
        fflush(stdout);

        if (fgets(command, sizeof(command), stdin) == NULL)
            break;

        command[strcspn(command, "\n")] = '\0';

        if (strcmp(command, "exit") == 0)
            break;

        pid_t pid = fork();

        if (pid < 0)
        {
            perror("fork");
            return 1;
        }
        else if (pid == 0)
        {
            printf("Child PID = %d\n", getpid());

            execlp(command, command, NULL);

            perror("exec");
            exit(1);
        }
        else
        {
            printf("Parent PID = %d\n", getpid());

            wait(NULL);
        }
    }
}
```

```

    else
    {
        printf("Parent PID = %d\n", getppid());

        wait(NULL);

        printf("Command completed.\n");
    }
}

return 0;
}
Lahari@Lahari:~/OS_Skill1$ gcc src/minishell.c -o bin/minishell
Lahari@Lahari:~/OS_Skill1$ ./bin/minishell
MiniShell> ls
Parent PID = 2085
Child PID = 2086
Makefile bin docs include sample.txt src trace.txt
Command completed.
MiniShell> pwd
Parent PID = 2085
Child PID = 2087
/home/lahari/OS_Skill1
Command completed.
MiniShell> date
Parent PID = 2085
Child PID = 2090
Wed Aug 12 07:06:11 UTC 2026
Command completed.
MiniShell> exit
Lahari@Lahari:~/OS_Skill1$ pwd
/home/lahari/OS_Skill1
Lahari@Lahari:~/OS_Skill1$ tree
Command 'tree' not found, but can be installed with:
sudo snap install tree # version 2.1.3+pkg-5852, or
sudo apt install tree # version 2.3.1-1
See 'snap info tree' for additional versions.
Lahari@Lahari:~/OS_Skill1$ sudo apt install tree -y
[sudo: authenticate] Password:
Installing:
  tree

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 107
  Download size: 53.5 kB
  Space needed: 125 kB / 1024 GB available

```

```
Get:1 http://archive.ubuntu.com/ubuntu resolute/universe amd64 tree amd64 2.3.1-1 [53.5 kB]
Fetched 53.5 kB in 2s (27.9 kB/s)
Selecting previously unselected package tree.
(Reading database ... 42298 files and directories currently installed.)
Preparing to unpack .../tree_2.3.1-1_amd64.deb ...
Unpacking tree (2.3.1-1) ...
Setting up tree (2.3.1-1) ...
Processing triggers for man-db (2.13.1-1build1) ...
Lahari@Lahari:~/OS_Skill1$ tree
.
├── Makefile
├── bin
│   ├── execl_demo
│   ├── execv_demo
│   ├── execvp_demo
│   ├── fork_demo
│   ├── hello
│   ├── minishell
│   ├── multiple_fork
│   ├── process_tree
│   ├── wait_demo
│   └── waitpid_demo
├── docs
├── include
├── sample.txt
├── src
│   ├── execl_demo.c
│   ├── execv_demo.c
│   ├── execvp_demo.c
│   ├── fork_demo.c
│   ├── hello.c
│   ├── minishell.c
│   ├── multiple_fork.c
│   ├── process_tree.c
│   ├── wait_demo.c
│   └── waitpid_demo.c
└── trace.txt

5 directories, 23 files
Lahari@Lahari:~/OS_Skill1$ git status
On branch master
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    Makefile
```

On branch master

Untracked files:

(use "git add <file>..." to include in what will be committed)

```
Makefile
bin/execl_demo
bin/execv_demo
bin/execvp_demo
bin/fork_demo
bin/minishell
bin/multiple_fork
bin/process_tree
bin/wait_demo
bin/waitpid_demo
src/execl_demo.c
src/execv_demo.c
src/execvp_demo.c
src/fork_demo.c
src/minishell.c
src/multiple_fork.c
src/process_tree.c
src/wait_demo.c
src/waitpid_demo.c
trace.txt
```

nothing added to commit but untracked files present (use "git add" to track)

```
lahari@Lahari:~/OS_Skill1$ git add .
```

```
lahari@Lahari:~/OS_Skill1$ git status
```

On branch master

Changes to be committed:

(use "git restore --staged <file>..." to unstage)

```
new file:   Makefile
new file:   bin/execl_demo
new file:   bin/execv_demo
new file:   bin/execvp_demo
new file:   bin/fork_demo
new file:   bin/minishell
```


nothing added to commit but untracked files present (use "git add" to track)

```
lahari@Lahari:~/OS_Skill1$ git add .
```

```
lahari@Lahari:~/OS_Skill1$ git status
```

On branch master

Changes to be committed:

(use "git restore --staged <file>..." to unstage)

```
new file:   Makefile
new file:   bin/execl_demo
new file:   bin/execv_demo
new file:   bin/execvp_demo
new file:   bin/fork_demo
new file:   bin/minishell
new file:   bin/multiple_fork
new file:   bin/process_tree
new file:   bin/wait_demo
new file:   bin/waitpid_demo
new file:   src/execl_demo.c
new file:   src/execv_demo.c
new file:   src/execvp_demo.c
new file:   src/fork_demo.c
new file:   src/minishell.c
new file:   src/multiple_fork.c
new file:   src/process_tree.c
new file:   src/wait_demo.c
new file:   src/waitpid_demo.c
new file:   trace.txt
```

```
lahari@Lahari:~/OS_Skill1$ git commit -m "Complete OS Skill 1"
```

```
new file:   trace.txt
lahari@Lahari:~/OS_Skill1$ git commit -m "Complete OS Skill 1"
[master 32c0d7b] Complete OS Skill 1
20 files changed, 359 insertions(+)
create mode 100644 Makefile
create mode 100755 bin/execl_demo
create mode 100755 bin/execv_demo
create mode 100755 bin/execvp_demo
create mode 100755 bin/fork_demo
create mode 100755 bin/minishell
create mode 100755 bin/multiple_fork
create mode 100755 bin/process_tree
create mode 100755 bin/wait_demo
create mode 100755 bin/waitpid_demo
create mode 100644 src/execl_demo.c
create mode 100644 src/execv_demo.c
create mode 100644 src/execvp_demo.c
create mode 100644 src/fork_demo.c
create mode 100644 src/minishell.c
create mode 100644 src/multiple_fork.c
create mode 100644 src/process_tree.c
create mode 100644 src/wait_demo.c
create mode 100644 src/waitpid_demo.c
create mode 100644 trace.txt
lahari@Lahari:~/OS_Skill1$ git status
On branch master
nothing to commit, working tree clean
lahari@Lahari:~/OS_Skill1$ git log --oneline
32c0d7b (HEAD -> master) Complete OS Skill 1
0791805 Initial OS Skill 1 project
```